

Shreve: Stochastic Calculus for Finance II
Key Results, Theorems, Definitions, etc.
Charles Zitella

Abstract

This document contains a summary of key results, theorems, and definitions that I deem important or interesting. This is not a comprehensive summary of the textbook, but a taste.

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1. GENERAL PROBABILITY THEORY

1.1. INFINITE PROBABILITY SPACES

Used to model a situation in which a random experiment with infinitely many possible outcomes is conducted. The problem with uncountably infinite sample spaces, like uniform sampling from the interval $[0, 1]$, is that for most experiments the probability of any particular outcome is 0.

Definition 1 (σ -algebra). Let Ω be a nonempty set, and $\mathcal{F}$ a collection of subsets of Ω . We say that $\mathcal{F}$ is a **σ -algebra** if the following holds:

- (1) $\emptyset \in \mathcal{F}$
- (2) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$
- (3) $A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$

Note that this also implies closed under finite unions and closed under countable intersections (since intersection is complement of union of complements). Additionally the whole space $\Omega = \emptyset^c \in \mathcal{F}$.

Definition 2 (probability measure). Let Ω nonempty set, $\mathcal{F}$ a σ -algebra of subsets of Ω . A **probability measure** $\mathbb{P}$ is a function that assigns a number in $[0, 1]$ to every set $A \in \mathcal{F}$ called the probability of A and written $\mathbb{P}(A)$. We require that

- (1) $\mathbb{P}(\Omega) = 1$
- (2) (countable additivity) If $A_1, A_2, \dots$ is a sequence of disjoint sets in $\mathcal{F}$, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **probability space**.

Example 1. *Uniform Lebesgue Measure on $[0, 1]$*

Define the probability of closed intervals $[a, b]$ to be $\mathbb{P}[a, b] = b - a$, $0 \leq a \leq b \leq 1$. Note if $a = b$ then $[a, b] = \{a\}$ and has a probability of 0. Note this also implies $\mathbb{P}(a, b) = \mathbb{P}[a, b] = b - a$.

All sets whose probabilities are determined by the defined formula and the properties of probability measures are exactly those belonging to the **Borel σ -algebra** $\mathcal{B}([0, 1])$, which is the σ -algebra generated by the set of open or closed sub-intervals. The sets in this σ -algebra are called **Borel sets**.

Example 2. *Infinite, Independent Coin Toss Space*

A coin is tossed infinitely many times and we let Ω_∞ be the set of possible outcomes. The probability of a head is $p > 0$ and of tails $q = 1 - p > 0$. We can define a probability measure that gives a probability to every set that can be described in terms of finitely many tosses, which in turn determines the probability of sets that are not describable with finitely many coin tosses. One can then show that every sequence in Ω_∞ has probability zero. This brings about the notion of *almost sure*.

Definition 3 (almost sure). in the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, if a set $A \in \mathcal{F}$ satisfies $\mathbb{P}(A) = 1$ then it occurs **almost surely**.

1.2. RANDOM VARIABLES AND DISTRIBUTIONS

Definition 4. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A **random variable** is a real valued function X defined on Ω with the property that for every Borel set $B \in \mathcal{B}(\mathbb{R})$, the subset of Ω given by

$$\{X \in B\} = \{\omega \in \Omega : X(\omega) \in B\}$$

is in the σ -algebra $\mathcal{F}$.

Example 3. *Stock Prices*

Consider the space of infinite, independent coin tosses from before $(\Omega_\infty, \mathcal{F}_\infty, \mathbb{P})$. Define stock prices using the formulas

$$\begin{aligned} S_0(\omega) &= 4, \\ S_1(\omega) &= \begin{cases} 8, & \omega_1 = H, \\ 2, & \omega_1 = T, \end{cases} \\ S_2(\omega) &= \begin{cases} 16, & \omega_1 = \omega_2 = H, \\ 4, & \omega_1 \neq \omega_2, \\ 1, & \omega_1 = \omega_2 = T, \end{cases} \\ S_{n+1}(\omega) &= \begin{cases} 2S_n(\omega), & \omega_{n+1} = H, \\ \frac{1}{2}S_n(\omega), & \omega_{n+1} = T. \end{cases} \end{aligned}$$

These are all random variables and thus we can compute the probability they take certain values, like $\mathbb{P}\{S_2 = 4\} = \mathbb{P}(A_{HT} \cup A_{TH}) = 2pq$.

Definition 5 (distribution measure). Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The **distribution measure** of X is the probability measure μ_X that assigns to each Borel subset $B \subset \mathbb{R}$ the mass $\mu_X(B) = \mathbb{P}\{X \in B\}$.

It's important to note that random variables and distributions are two different concepts. Two different random variables can have the same distribution and a single random variable can have two different distributions.

Example 4. Distributions

Consider the uniform measure on $[0, 1]$ defined before, and call it $\mathbb{P}$. Now consider $X(\omega) = \omega$, and $Y(\omega) = 1 - \omega$. These are two distinct RVs, both defined on $\Omega = [0, 1]$, but they share the same distribution:

$$\mu_Y[a, b] = \mathbb{P}\{Y \in [a, b]\} = \mathbb{P}\{\omega : a \leq 1 - \omega \leq b\} = \mathbb{P}[1 - b, 1 - a] = (1 - a) - (1 - b) = b - a = \mu_X[a, b]$$

Note we can also define a probability measure on $[0, 1]$ such that the distribution of X and Y are different. One such is example is the following:

$$\tilde{\mathbb{P}}[a, b] = \int_a^b 2\omega \, d\omega = b^2 - a^2, \quad 0 \leq a \leq b \leq 1$$

Once the distribution measure is defined for all closed intervals $[a, b] \subset \mathbb{R}$, it is determined for every Borel subset of $\mathbb{R}$.

We can also record the distribution of an RV in terms of its **cumulative distribution function** (cdf), which is defined as $F(x) = \mathbb{P}\{X \leq x\} = \mu_X(-\infty, x]$, $x \in \mathbb{R}$. The cdf and the distribution measure essentially give the same amount of information, but we can have more detail in two special cases. Firstly if there's a nonnegative **density function** $f(x)$ such that

$$\mu_X[a, b] = \mathbb{P}\{a \leq X \leq b\} = \int_a^b f(x) \, dx \quad -\infty < a \leq b < \infty$$

Second if there's a **probability mass function**. In this case there exists a possibly infinite sequence $x_1, x_2, \dots$ such that the rv X takes a value in the sequence with probability one. In this case $p_i = \mathbb{P}\{X = x_i\}$ The mass assigned to a borel set $B \subset \mathbb{R}$ is

$$\mu_X(B) = \sum_{i: x_i \in B} p_i \quad B \in \mathcal{B}(\mathbb{R})$$

Example 5. Standard Normal Random Variable

Let

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

be the standard normal density, and define the cumulative normal distribution function

$$N(x) = \int_{-\infty}^x \varphi(\zeta) \, d\zeta$$

Note that this is strictly increasing and continuous, so it has a strictly increasing inverse $N^{-1}(y)$. We can construct an rv with a normal distribution on any probability space $(\Omega, \mathcal{F}, \mathbb{P})$ by starting with a uniformly distributed rv, Y and setting $X = N^{-1}(Y)$. Then

$$\begin{aligned} \mu_X[a, b] &= \mathbb{P}\{\omega \in \Omega : a \leq X(\omega) \leq b\} \\ &= \mathbb{P}\{\omega \in \Omega : a \leq N^{-1}(Y(\omega)) \leq b\} \\ &= \mathbb{P}\{\omega \in \Omega : N(a) \leq Y(\omega) \leq N(b)\} \\ &= N(b) - N(a) \\ &= \int_a^b \varphi(x) \, dx \end{aligned}$$

This μ_X is called the **standard normal distribution**, and any rv with this distribution regardless of probability space is called a standard normal rv.

1.3. EXPECTATIONS

We want to define a notion of the "average value" of an rv X on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. If Ω is finite, we can expect the "average" of the rv X to be

$$\mathbb{E}X = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega)$$

If Ω is countably infinite, we define $\mathbb{E}X$ as the same sum but infinite, but what about continuous rvs? This requires the Lebesgue integral, which I will not go into the construction of here (Analysis 3).

Theorem 1 (Properties of the Lebesgue Integral). Let X be a random variable on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- (1) If X takes on only finitely many values $y_1, y_2, \dots, y_N$, then

$$\int_{\Omega} X(\omega) \mathbb{P}(\omega) = \sum_{k=1}^N y_k \mathbb{P}\{X = y_k\}$$

- (2) (integrability) The r.v. X is integrable iff $\int_{\Omega} |X(\omega)| d\mathbb{P}(\omega) < \infty$

- (3) (comparison) If $X \leq Y$ almost surely,¹ and if $\int_{\Omega} X(\omega) d\mathbb{P}(\omega)$ and $\int_{\Omega} Y(\omega) d\mathbb{P}(\omega)$ are defined, then

$$\int_{\Omega} X(\omega) d\mathbb{P}(\omega) \leq \int_{\Omega} Y(\omega) d\mathbb{P}(\omega)$$

If $X = Y$ almost surely and these integrals are defined, then they are equal.

- (4) (linearity) If $\alpha, \beta \in \mathbb{R}_{\geq 0}$, X, Y nonnegative and integrable, then

$$\int_{\Omega} (\alpha X(\omega) + \beta Y(\omega)) d\mathbb{P}(\omega) = \alpha \int_{\Omega} X(\omega) d\mathbb{P}(\omega) + \beta \int_{\Omega} Y(\omega) d\mathbb{P}(\omega)$$

Remark. Let $A \in \mathcal{F}$, then

$$\int_A X(\omega) d\mathbb{P}(\omega) = \int_{\Omega} \mathbb{I}_A(\omega) X(\omega) d\mathbb{P}(\omega)$$

Definition 6 (expectation). Let X be a random variable on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The **expectation** (or expected value) of X is defined to be

$$\mathbb{E}X = \int_{\Omega} X(\omega) d\mathbb{P}(\omega)$$

This definition makes sense if X is integrable, i.e. $\mathbb{E}|X| < \infty$ or $X \geq 0$ almost surely.

Since the expectation is defined in terms of a Lebesgue integral, all the previous properties mentioned for the Lebesgue integral also hold for the expectation as well as **Jensen's inequality**, which states that if φ is a convex, real-valued function defined on $\mathbb{R}$, and if $\mathbb{E}|X| < \infty$, then $\varphi(\mathbb{E}X) \leq \mathbb{E}\varphi(X)$.

Example 6. *Counting Heads*

Consider the infinite coin toss space defined previously, and let $\mathbb{P}$ be the probability measure that corresponds to probability $\frac{1}{2}$ for heads on each toss. Let

$$Y_n(\omega) = \begin{cases} 1 & \text{if } \omega_n = H \\ 0 & \text{if } \omega_n = T \end{cases}$$

Then

$$\mathbb{E}Y_n = 1 \cdot \mathbb{P}\{Y_n = 1\} + 0 \cdot \mathbb{P}\{Y_n = 0\} = \frac{1}{2}$$

¹i.e. $\mathbb{P}\{X \leq Y\} = 1$

Example 7. *Rationals and Irrationals*

Consider the probability space with $\Omega = [0, 1]$ and $\mathbb{P}$ the lebesgue measure. Define

$$X(\omega) = \begin{cases} 1 & \text{if } \omega \text{ is irrational} \\ 0 & \text{if } \omega \text{ is rational} \end{cases}$$

Then

$$\mathbb{E}X = 1 \cdot \mathbb{P}\{\omega \in [0, 1] : \omega \text{ is irrational}\} + 0 \cdot \mathbb{P}\{\omega \in [0, 1] : \omega \text{ is rational}\}$$

Note there are only countably many rational numbers, and the probability of each individual one is 0, and thus by countable additivity $\mathbb{P}\{\omega \in [0, 1] : \omega \text{ is rational}\} = 0$. Therefore the probability of the irrationals must be 1, and subsequently $\mathbb{E}X = 1$.

It's worth noting that the lebesgue integral is essential here, because the integral would be undefined with the Riemann definition.

Definition 7 (Lebesgue measure). The **Lebesgue measure** on $\mathbb{R}$, $\mathcal{L}$, is a measure that satisfies countable additivity and is defined as $\mathcal{L}[a, b] = b - a$ whenever $a \leq b$.

Definition 8 (almost everywhere). We say that a property holds **almost everywhere** (a.e.) if the set of numbers that fail to satisfy the property is of Lebesgue measure 0.

Theorem 2 (Riemann vs. Lebesgue integrals). Letting f be a bounded function defined on $\mathbb{R}$, and $a < b$ in $\mathbb{R}$,

- (1) The Riemann integrable $\int_a^b f(x) dx$ exists iff f is continuous almost everywhere.
- (2) If the Riemann integral is defined, then f is Borel measurable² and the Riemann and Lebesgue integrals are equal.

1.4. CONVERGENCE OF INTEGRALS

Definition 9 (convergence of random variables). Let $X_1, X_2, \dots$ be a sequence of random variables, all defined on the same probability space. We say $\{X_k\}_{k \geq 0}$ **converges to X almost surely** if

$$\mathbb{P}\left\{\omega \in \Omega : \lim_{k \rightarrow \infty} X_k(\omega) = X(\omega)\right\} = 1$$

Example 8. *Strong Law of Large Numbers*

Consider the infinite coin toss space with $\mathbb{P}$ chosen such that the probability of each head is $\frac{1}{2}$. Let

$$Y_k = \begin{cases} 1 & \text{if } \omega_k = H \\ 0 & \text{if } \omega_k = T \end{cases}$$

and

$$H_n = \sum_{k=1}^n Y_k$$

Then H_n is the number of heads in the first n tosses, and the strong law of large number asserts that

$$\lim_{n \rightarrow \infty} \frac{H_n}{n} = \frac{1}{2} \quad \text{almost surely}$$

Almost surely because there are some sequences where this isn't true, like all heads, but all those sequences have measure 0.

²The preimage of every Borel set is a Borel set

Definition 10 (convergence of measurable functions). If $\{f_k\}_{k \geq 1}$ is a sequence of real valued, Borel measurable functions on $\mathbb{R}$, then the sequence **converges to f almost everywhere** if

$$\mathcal{L} \left\{ x \in \mathbb{R} : \lim_{k \rightarrow \infty} f_k(x) \neq f(x) \right\} = 0$$

Almost everywhere and almost surely convergence is basically the same thing, just almost surely is used with r.v.s and probability spaces.

Theorem 3 (monotone convergence). If $\{X_n\}_{n \geq 0}$ is a sequence of almost surely monotone increasing random variables that converge almost surely, then

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} X_n(\omega) d\mathbb{P}(\omega) = \int_{-\infty}^{\infty} \lim_{n \rightarrow \infty} X_n(\omega) d\mathbb{P}(\omega)$$

Therefore $\lim_{n \rightarrow \infty} \mathbb{E}X_n = \mathbb{E}X$ where $\lim_{n \rightarrow \infty} X_n = X$.

Monotone convergence theorem allows us to extend the expectation definition for the discrete case to countably infinite cases, getting that if X takes on countably many values then $\mathbb{E}X = \sum_{k=0}^{\infty} x_k \mathbb{P}\{X = x_k\}$

Example 9. *St Petersburg paradox*

Same infinite coin toss space, define

$$X(\omega) = \begin{cases} 2 & \text{if } \omega_1 = H \\ 4 & \text{if } \omega_1 = T, \omega_2 = H \\ 8 & \text{if } \omega_1 = T, \omega_2 = T, \omega_3 = H \\ \vdots & \\ 2^k & \text{if } \omega_1 = \omega_2 = \dots = \omega_{k-1} = T, \omega_k = H \end{cases}$$

Then this is finite almost surely, since only non finite case is all tails. Then

$$\mathbb{E}X = \sum_{k \geq 1} 2^k \mathbb{P}\{X = 2^k\} = \sum_{k \geq 1} 2^k \frac{1}{2^k} = \infty$$

Theorem 4 (dominated convergence). If $\{X_n\}_{n \geq 0}$ is a sequence of random variables and there exists a rv Y such that $\mathbb{E}Y < \infty$ and $|X_n| \leq Y$ almost surely for all n , then we can bring the limit inside the integral, similar to the MCT.

1.5. COMPUTING EXPECTATIONS

Computing expectations using the integral definition requires us to somehow turn it into an integral on $\mathbb{R}$. This can be done using a density function, but if we don't have one we can use the following. Note that the distribution measure is a probability measure on $\mathbb{R}$.

Theorem 5. X a rv on probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and g a Borel measurable function on $\mathbb{R}$. Then

$$\mathbb{E}|g(X)| = \int_{\mathbb{R}} |g(x)| d\mu_X(x)$$

and if this quantity is finite then

$$\mathbb{E}g(X) = \int_{\mathbb{R}} g(x) d\mu_X(x)$$

Proof. The proof requires first showing that it is true for indicator functions, and then building up through simple functions then non-negative borel measurable functions and finally all Borel measurable functions. $\square$

We still can't compute this, but there are several cases we can. If X takes only finitely many values then μ_X places a mass of size $p_k = \mathbb{P}\{X = x_k\}$ at each x_k so we get the formula

$$\mathbb{E}g(X) = \int_{\mathbb{R}} g(x) d\mu_X(x) = \sum_{k=0}^n g(x_k)p_k$$

More commonly we have a nonnegative, Borel measurable density function f on $\mathbb{R}$ such that

$$\mu_X(B) = \int_B f(x) dx \quad \text{for every Borel measurable set } B$$

and in the case when f is bounded and a.e. continuous, we can compute the Riemann integral and compute the expectation.

Theorem 6. X a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ and g Borel measurable on $\mathbb{R}$. If X has a density then

$$\mathbb{E}|g(X)| = \int_{-\infty}^{\infty} |g(x)|f(x) dx$$

and if this is finite then

$$\mathbb{E}g(X) = \int_{-\infty}^{\infty} g(x)f(x) dx$$

Proof. Once again, show true for indicator functions and build your way up to all Borel measurable functions. $\square$

1.6. CHANGE OF MEASURE

Theorem 7. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a prob. space and Z an almost surely nonnegative random variable with $\mathbb{E}Z = 1$. For $A \in \mathcal{F}$, define

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega)$$

Then $\tilde{\mathbb{P}}$ is a probability measure, and if X is a nonnegative random variable then

$$\tilde{\mathbb{E}}X = \mathbb{E}[XZ]$$

If Z is almost surely strictly positive then we also have

$$\mathbb{E}X = \tilde{\mathbb{E}}\left[\frac{X}{Z}\right]$$

Remark. For a rv with positive and negative values we can apply the theorem to its positive and negative parts provided the resulting subtraction does not result in $\infty - \infty$.

Proof of theorem. First

$$\tilde{\mathbb{P}}(\Omega) = \int_{\Omega} Z(\omega) d\mathbb{P}(\omega) = \mathbb{E}Z = 1$$

and for disjoint $A_1, A_2, \dots \in \mathcal{F}$, since $B_n = \cup_{k=1}^n A_k$ is increasing and $\lim_{n \rightarrow \infty} B_n = B_{\infty} = \cup_{k=1}^{\infty} A_k$ we get by monotone convergence theorem

$$\tilde{\mathbb{P}}\left(\bigcup_{k=1}^{\infty} A_k\right) = \int_{\Omega} \mathbb{I}_{B_{\infty}}(\omega) Z(\omega) d\mathbb{P}(\omega) = \lim_{n \rightarrow \infty} \int_{\Omega} \mathbb{I}_{B_n}(\omega) Z(\omega) d\mathbb{P}(\omega) = \sum_{k=1}^{\infty} \int_{\Omega} \mathbb{I}_{A_k}(\omega) Z(\omega) d\mathbb{P}(\omega) = \sum_{k=1}^{\infty} \tilde{\mathbb{P}}(A_k)$$

Thus it is a probability measure, also for the second part we prove for indicators and move our way up, here is the proof for indicators, assume $X = \mathbb{I}_A$

$$\tilde{\mathbb{E}}X = \tilde{\mathbb{P}}(A) = \int_{\Omega} \mathbb{I}_A(\omega) Z(\omega) d\mathbb{P}(\omega) = \mathbb{E}[\mathbb{I}_A Z] = \mathbb{E}[XZ]$$

$\square$

Definition 11. Two probability measures on the same space $(\Omega, \mathcal{F})$ are said to be **equivalent** if they agree on which sets have probability zero.

The previously defined $\mathbb{P}$ and $\tilde{\mathbb{P}}$ are equivalent.

Example 10. Refer to example 4, our space is $[0, 1]$, $\mathbb{P}$ is the Lebesgue measure, and

$$\tilde{\mathbb{P}}[a, b] = \int_a^b 2\omega \, d\omega = b^2 - a^2, \quad 0 \leq a \leq b \leq 1$$

Then since $\mathbb{P}$ is the lebesgue measure, integrating with respect to ω is the same thing as integrating with respect to $\mathbb{P}(\omega)$, so we get

$$\tilde{\mathbb{P}}[a, b] = \int_a^b 2\omega \, d\mathbb{P}(\omega)$$

and since this holds for every closed interval it holds for every Borel set, and thus for $B \in \mathcal{B}$

$$\tilde{\mathbb{P}}(B) = \int_B 2\omega \, d\mathbb{P}(\omega)$$

Then with $Z(\omega) = 2\omega$,

$$\mathbb{E}Z = \int_0^1 2\omega \, d\omega = 1$$

So we can apply the previous theorem to any non-negative RV $X(\omega)$ and get

$$\int_0^1 X(\omega) \, d\tilde{\mathbb{P}}(\omega) = \tilde{\mathbb{E}}X = \mathbb{E}[XZ] = \int_0^1 X(\omega) \cdot 2\omega \, d\mathbb{P}(\omega)$$

and thus

$$d\tilde{\mathbb{P}}(\omega) = 2\omega \, d\mathbb{P}(\omega)$$

Definition 12 (Radon-Nikodym derivative). $(\Omega, \mathcal{F}, \mathbb{P})$ a prob. space and $\tilde{\mathbb{P}}$ another equivalent prob. measure on the same space, with an almost surely positive rv Z such that $\tilde{\mathbb{P}}(A) = \int_A Z(\omega) \, d\mathbb{P}(\omega)$. Then Z is called the **Radon-Nikodym derivative** of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$, and we write

$$Z = \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}$$

Theorem 8 (Radon-Nikodym). $\mathbb{P}$ and $\tilde{\mathbb{P}}$ equivalent probability measures on the same space $(\Omega, \mathcal{F})$. Then there exists an almost surely positive random variable Z such that $\mathbb{E}Z = 1$ and for every $A \in \mathcal{F}$

$$\tilde{\mathbb{P}}(A) = \int_A Z(\omega) \, d\mathbb{P}(\omega)$$

1.7. EXERCISES

2. INFORMATION AND CONDITIONING